Excretory Products and their Elimination

16.0 Why Excretion Exists - The Chapter's Starting Point

Animals **accumulate** ammonia, urea, uric acid, CO_2, water and ions like Na^+, K^+, Cl^-, **phosphate, sulphate**, etc. They pick these up in two ways **- either by metabolic activities** or **by other means like excess ingestion**. These substances then **have to be removed totally or partially**.

> ⓘ ***[NOTE]*** *This chapter is about the **mechanisms of elimination** of these substances, with **special emphasis on the common nitrogenous wastes**.*

Ammonia, urea and uric acid are the **major forms of nitrogenous wastes** excreted by animals. The three differ in **toxicity**, and toxicity is what decides **how much water** an animal must spend to get rid of them:

Waste	Toxicity	Water needed to eliminate it
Ammonia	**Most toxic** form	**Large amount** of water required
Urea	Intermediate	Intermediate
Uric acid	**Least toxic**	Removed with a **minimum loss** of water

> ☆ ***[MEMORY AID - not in NCERT] Toxicity falls, water saved rises -*** *Ammonia > Urea > Uric acid in toxicity, so the **same order reversed** gives water economy. An animal's choice of waste is really a statement about **how much water it can afford to lose**.*

Because of that trade-off, animals are classified by **which nitrogenous waste they excrete**. All three names are built the same way, on the waste plus -*telism*:

16.0a Ammonotelism

- **Ammonotelism -** the **process of excreting ammonia**.
  - **Who:** many **bony fishes, aquatic amphibians and aquatic insects** are **ammonotelic** in nature.
  - **How:** ammonia is **readily soluble**, so it is generally excreted **by diffusion across body surfaces** or **through gill surfaces (in fish)** as **ammonium ions**.
  - **Kidneys do not play any significant role** in ammonia's removal.

16.0b Ureotelism

Terrestrial adaptation necessitated the production of **lesser toxic** nitrogenous wastes like **urea and uric acid**, for **conservation of water**.

- **Ureotelic animals -** those that **mainly excrete urea**: **mammals, many terrestrial amphibians and marine fishes**.
  - **Where urea is made:** ammonia produced by metabolism is **converted into urea in the liver** of these animals, and **released into the blood**, which is then **filtered and excreted out by the kidneys**.
  - **Some urea is deliberately kept:** some amount of urea **may be retained in the kidney matrix** of some of these animals **to maintain a desired osmolarity**.

16.0c Uricotelism

- **Uricotelic animals - reptiles, birds, land snails and insects**, which excrete nitrogenous wastes as **uric acid in the form of pellet or paste**, with a **minimum loss of water**.

> ⓘ ***[NOTE] Osmoregulation*** *(assumed by Exercise 8; the NCERT body uses the word but never defines it): the **maintenance of the body's water and electrolyte balance -** that is, keeping the **volume and ionic concentration of body fluids** within a narrow range. Watch how often the structures below are described as doing **this** job rather than a waste-removal job; in many invertebrates the two are the same organ's work.*

16.0d Excretory Structures Across the Animal Kingdom

A **survey of the animal kingdom** presents a **variety of excretory structures**. In **most invertebrates** these structures are **simple tubular forms**, whereas **vertebrates have complex tubular organs called kidneys**. Some of these structures are:

Structure	Found in	What it does
Protonephridia or **flame cells**	**Platyhelminthes** (Flatworms, e.g., *Planaria*), **rotifers**, some **annelids**, and the cephalochordate - **Amphioxus**	Primarily **ionic and fluid volume regulation**, i.e., **osmoregulation**
Nephridia	**Earthworms** and other **annelids**	Remove **nitrogenous wastes** and maintain a **fluid and ionic balance**
Malpighian tubules	Most **insects**, including **cockroaches**	Removal of **nitrogenous wastes** and **osmoregulation**
Antennal glands or **green glands**	**Crustaceans** like **prawns**	Perform the **excretory function**

☆ ***[MEMORY AID - not in NCERT] Protonephridia are the odd one out.*** *Nephridia and Malpighian tubules are named as doing **both** jobs (wastes + balance); protonephridia are named as **primarily osmoregulatory**. A NEET stem that asks which structure is 'primarily concerned with ionic and fluid volume regulation' is asking for **flame cells**.*

Chapter contents (p. 205 margin panel):

16.1 Human Excretory System

In humans, the excretory system consists of **four parts** (Figure 16.1):

- a **pair of kidneys**
- **one pair of ureters**
- a **urinary bladder**
- a **urethra**

16.1a The Kidney - Position, Size and Outer Features

Kidneys are reddish brown, bean shaped structures. They sit **between the levels of the last thoracic and third lumbar vertebra**, **close to the dorsal inner wall of the abdominal cavity**.

Dimensions of one adult human kidney:

Length	Width	Thickness	Average weight
10-12 cm	**5-7 cm**	**2-3 cm**	**120-170 g**

☆ ***[MEMORY AID - not in NCERT] 10-12 / 5-7 / 2-3, then 120-170 g.*** *The three dimensions run **downwards in pairs** and the weight is the only three-digit number in the set. NEET has asked for each of these four figures directly.*

Towards the **centre of the inner concave surface** of the kidney is a **notch called the hilum**, through which the **ureter, blood vessels and nerves enter**. **Inner to the hilum** is a **broad funnel shaped space called the renal pelvis**, with **projections called calyces**.

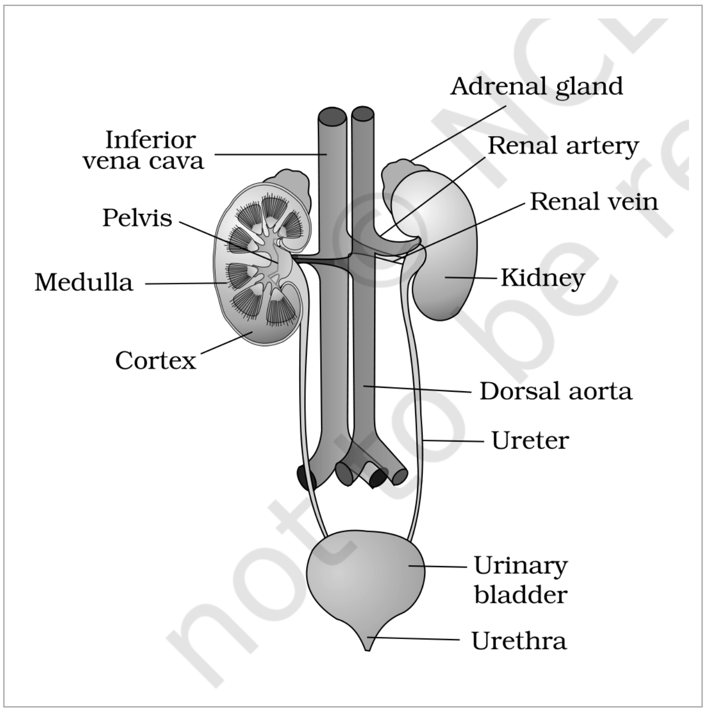

Figure 16.1 Human Urinary system

> ⓘ ***[NOTE] Figure 16.1 callouts, verbatim:*** *Inferior vena cava; Adrenal gland; Renal artery; Renal vein; Pelvis; Kidney; Medulla; Cortex; Dorsal aorta; Ureter; Urinary bladder; Urethra.*

> ⓘ ***[NOTE] Two vessels the figure names but the prose never does.*** *Figure 16.1 draws the **renal artery** branching off the **dorsal aorta**, and the **renal vein** draining into the **inferior vena cava**. The running text of the chapter names neither the dorsal aorta nor the inferior vena cava, so read them off the figure: **aorta -> renal artery -> kidney -> renal vein -> inferior vena cava**. The figure also shows the **adrenal gland** capping each kidney.*

16.1b Inside the Kidney - Cortex, Medulla and the Columns

- The **outer layer** of the kidney is a **tough capsule**.
- Inside, there are **two zones -** an **outer cortex** and an **inner medulla**.
- The **medulla is divided into a few conical masses -** the **medullary pyramids - projecting into the calyces** (sing.: **calyx**).
- The **cortex extends in between the medullary pyramids** as **renal columns** called **Columns of Bertini** (Figure 16.2).

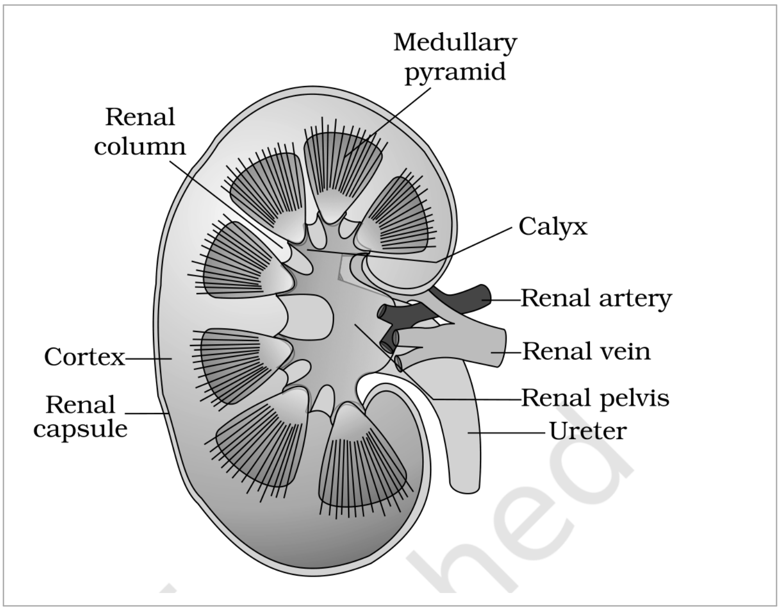

Figure 16.2 Longitudinal section (Diagrammatic) of Kidney

> ⓘ ***[NOTE] Figure 16.2 callouts, verbatim:*** *Medullary pyramid; Renal column; Calyx; Renal artery; Renal vein; Renal pelvis; Ureter; Cortex; Renal capsule.*

> ☆ ***[MEMORY AID - not in NCERT] Columns of Bertini are cortex, not medulla.*** *They are the cortex reaching* ***inwards between*** *the pyramids. A stem describing 'medullary tissue extending into the cortex' has the direction backwards.*

16.1c The Nephron - the Functional Unit

Each kidney has **nearly one million** complex tubular structures called **nephrons** (Figure 16.3), which are the **functional units**. Each nephron has **two parts** - the **glomerulus** and the **renal tubule**.

- **Glomerulus** - a **tuft of capillaries** formed by the **afferent arteriole**, a **fine branch of the renal artery**. Blood from the glomerulus is **carried away by an efferent arteriole**.
- **Bowman's capsule** - the **double walled cup-like structure** the renal tubule **begins with**, which **encloses the glomerulus**.
- **Malpighian body** or **renal corpuscle** - the **glomerulus alongwith Bowman's capsule**, taken together (Figure 16.4).

From Bowman's capsule onward, the tubule runs through four named regions in order:

- **Proximal convoluted tubule (PCT) -** the tubule continues from the capsule to form this **highly coiled network**.
- **Henle's loop -** the **next part**, **hairpin shaped**, with a **descending and an ascending limb**.
- **Distal convoluted tubule (DCT) -** the **ascending limb continues** as this **another highly coiled tubular region**.
- **Collecting duct -** the **DCTs of many nephrons open into** this **straight tube**. **Many collecting ducts converge** and **open into the renal pelvis through the medullary pyramids in the calyces**.

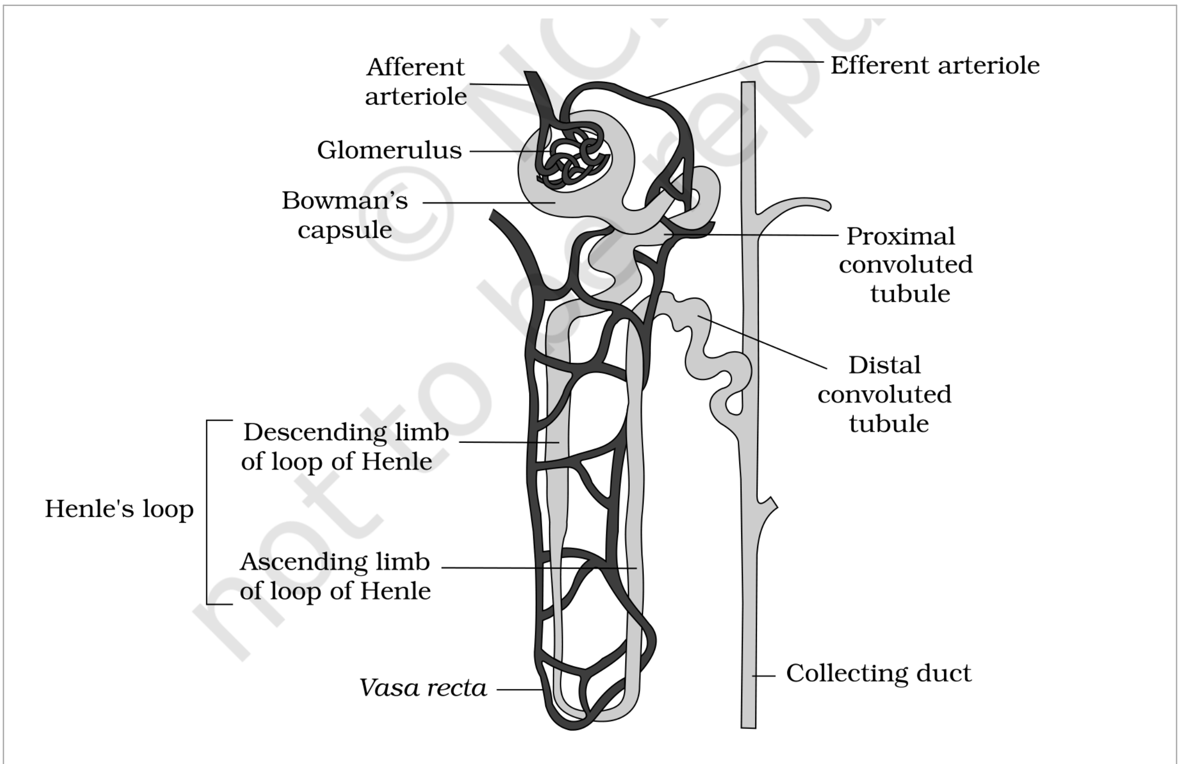

Figure 16.3 A diagrammatic representation of a nephron showing blood vessels, duct and tubule

> ⓘ ***[NOTE] Figure 16.3 callouts, verbatim:*** *Afferent arteriole; Efferent arteriole; Glomerulus; Bowman's capsule; Proximal convoluted tubule; Distal convoluted tubule; Descending limb of loop of Henle; Ascending limb of loop of Henle; Henle's loop; Vasa recta; Collecting duct.*

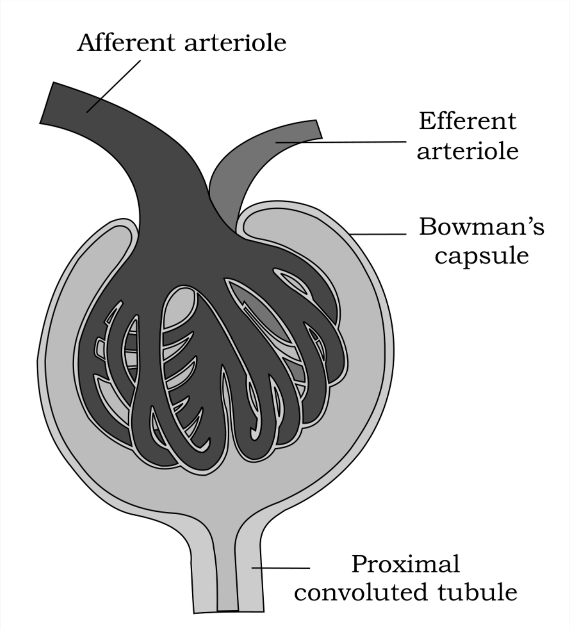

Figure 16.4 Malpighian body (renal corpuscle)

> ⓘ ***[NOTE] Figure 16.4 callouts, verbatim:*** *Afferent arteriole; Efferent arteriole; Bowman's capsule; Proximal convoluted tubule.*

16.1d Where the Parts Sit, and the Two Types of Nephron

The nephron is not evenly spread between the two zones. The **Malpighian corpuscle, PCT and DCT** are situated in the **cortical region** of the kidney, whereas the **loop of Henle dips into the medulla**.

How far the loop dips is what separates the **two types of nephron**:

	Cortical nephrons	Juxta medullary nephrons
How many	The **majority** of nephrons	**Some** of the nephrons
Loop of Henle	**Too short** - extends **only very little** into the medulla	**Very long** - runs **deep into the medulla**
Vasa recta	**Absent or highly reduced**	**Present**

> ☆ ***[MEMORY AID - not in NCERT] Juxta = next to.*** *A **juxta medullary** nephron sits **next to the medulla** and so can afford a **long** loop; the **cortical** nephron stays up in the cortex with a **short** loop and **no real vasa recta**. Remember which one loses the vasa recta **-** it is the **cortical** one, and 16.4 will show why that matters for concentrating urine.*

The blood supply follows the tubule. The **efferent arteriole** emerging from the glomerulus forms a **fine capillary network around the renal tubule**, called the **peritubular capillaries**. A **minute vessel of this network** runs **parallel to Henle's loop**, forming a **'U' shaped vasa recta**. **Vasa recta is absent or highly reduced in cortical nephrons.**